



INSIGHTSMASTERY ACADEMY
LEARNING HANDBOOK

Power Query Transform & Data Cleaning

Clean and shape data from CSV • Excel • Folder • SQL Server
Practical steps, patterns, and best practices

From Get Data → Transform Data → Close & Apply
Build reliable, refreshable queries

Power BI Desktop • Power Query Editor • M
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1. Power Query Editor Overview

Power Query is where you **shape and clean** data before it enters the Power BI model. Every change becomes an **Applied Step** that runs again on every refresh.

Key areas of the Editor

- **Ribbon** — Home, Transform, Add Column, View tabs with all transformation commands
- **Queries pane** (left) — list of queries; right-click to rename, duplicate, reference
- **Data preview** (center) — sample of rows; column quality / distribution when profiling is on
- **Applied Steps** (right) — history of every transformation; click a step to edit or delete
- **Formula bar** — shows the M expression for the selected step

■ Turn on profiling

View → Data Preview → enable Column quality, Column distribution, Column profile. By default profiling uses the top 1,000 rows; switch to entire dataset when needed.



2. Connect & Open Transform Data

Always prefer **Transform Data** over Load so you can clean before the model is built.

CSV / Text

Home → Get data → Text/CSV → select file → in the preview dialog set delimiter & encoding → **Transform Data**. Fix headers and types in the Editor.

Excel

Home → Get data → Excel workbook → select file → Navigator → check tables/sheets → **Transform Data**. Prefer Excel Tables over raw sheets.

Folder

Home → Get data → More → Folder → path → **Combine & Transform Data**. Pick a sample file/object; Power Query creates a combined query you then clean once for all files.

SQL Server

Home → Get data → SQL Server → server/database → Import or DirectQuery → Navigator → select tables → **Transform Data**. Keep foldable steps early for performance.



3. Essential Cleaning Transformations

Columns

- **Remove Columns** — select columns → Home/Transform → Remove Columns (or Remove Other Columns to keep selection).
- **Rename** — double-click header or right-click → Rename. Use clear business names.
- **Change Type** — select column → Transform → Data Type (or click type icon in header). Do this early.
- **Replace Values** — right-click column → Replace Values (standardize codes, fix typos).
- **Replace Errors** — right-click → Replace Errors (turn error cells into null or a default).

Rows

- **Remove Top/Bottom Rows** — Home → Remove Rows (junk titles, totals).
- **Remove Blank Rows** — Home → Remove Rows → Remove Blank Rows.
- **Remove Duplicates** — select key columns → Home → Remove Rows → Remove Duplicates.
- **Filter Rows** — column filter dropdown → uncheck nulls or unwanted values; or use Text Filters / Number Filters.
- **Keep Rows** — Keep Top N, Keep Range, Keep Duplicates when you need a subset.

Text cleanup

- **Trim** — Transform → Format → Trim (leading/trailing spaces).
- **Clean** — Transform → Format → Clean (non-printable characters).
- **Lowercase / Uppercase / Capitalize Each Word** — Transform → Format.



4. Source-Specific Cleaning Patterns

CSV — typical cleanup path

Step 1: Headers & types

Promote headers if needed (Home → Use First Row as Headers).
Set Date columns to Date, amounts to Decimal Number or Fixed Decimal Number.
If dates look wrong, use Data Type → Using Locale...

Step 2: Text & nulls

Trim key text columns (Customer Name, City).
Replace nulls in measures with 0 if business rules allow, or leave null.
Remove empty rows and any trailer rows from the export.

Excel — typical cleanup path

Step 1: Structure

If the sheet has title rows, Remove Top Rows until headers are on row 1, then Use First Row as Headers.
Remove total/subtotal rows (filter or Remove Bottom Rows).
Prefer connecting to a named Excel Table to avoid layout drift.

Step 2: Columns

Remove helper columns not needed in the model.
Split "City, State" with Split Column by Delimiter if required.
Standardize category values with Replace Values.

Folder — after Combine

After Combine & Transform Data you get one combined table. Clean that query once:

- Confirm Source.Name (or similar) column if you need file name as attribute; rename it.
- Set types, trim, replace values as for a single CSV/Excel.
- Filter out unwanted files earlier (Transform the file list before combining) if some files differ.

SQL Server — keep folding in mind

- Select only needed columns early (removes columns from the SELECT).
- Filter rows early so filters become WHERE clauses.
- Avoid early non-foldable steps (complex text transforms, index columns) when the table is large.
- Right-click a step → View Native Query to confirm folding is still active.



5. Unpivot, Fill, Split & Conditional Columns

Unpivot (wide → tall)

When months or categories are separate columns (Jan, Feb, Mar...), select the columns to unpivot (or select the ones to keep and choose Unpivot Other Columns). Transform → Unpivot Columns. You get Attribute + Value; rename them (e.g. Month, Sales) and set types.

Fill Down / Fill Up

For hierarchical exports where a category appears only on the first row of a group: select the column → Transform → Fill → Down (or Up).

Split Column

Transform → Split Column → By Delimiter / By Number of Characters / By Positions. Useful for “Last, First” names or codes embedded in text.

Conditional Column

Add Column → Conditional Column. Build simple if-then-else rules without writing M. For more complex logic use Add Column → Custom Column and an M expression.

```
// Example custom column (M)
if [Amount] >= 1000 then "High" else if [Amount] >= 100 then "Medium" else "Low"
```



6. Data Types, Locale & Query Folding

Data types that matter most

Use	For
Date	Calendar dates (no time)
Date/Time	Timestamps
Whole Number	IDs, counts, keys
Fixed Decimal Number	Currency / money (avoids float issues)
Decimal Number	General decimals
Text	Names, codes, categories
True/False	Flags

Table 1 — Common data types

Query folding (SQL Server especially)

Folding means Power Query pushes filters, selects, and joins to the source. Keep foldable steps at the top. Non-foldable steps (many text operations, unpivot, custom functions) force local processing of more data.

■ Check folding

Right-click an Applied Step → View Native Query. If the option is disabled, folding has already broken above that step.



7. Applied Steps & Maintainable Queries

- Name steps clearly (e.g. “Filtered cancelled orders”, not “Filtered Rows1”).
- Delete or reorder steps carefully — later steps may depend on earlier ones.
- Use Reference (right-click query → Reference) to branch a cleaned staging query into multiple outputs.
- Duplicate a query only when you truly need an independent copy.
- Document complex logic with comments in the Advanced Editor (`//` comment).
- Group related queries in folders in the Queries pane.

■ Staging pattern

Create a staging query that only cleans (types, trim, filter). Then create referenced queries that shape dimensions and facts from the staging query. One place to fix source issues.



8. End-to-End Cleaning Checklist

After opening Transform Data

- Headers correct (Use First Row as Headers if needed)
- Junk top/bottom rows removed
- Blank rows and obvious error rows removed
- Only required columns kept
- Columns renamed to business-friendly names
- Data types set (especially Date and numeric)
- Text trimmed and cleaned where needed
- Values standardized (Replace Values)
- Wide data unpivoted if required
- Duplicates handled according to rules
- Query renamed (e.g. FactSales, DimProduct)
- Applied Steps named clearly

Before Close & Apply

- Preview looks correct on sample and (if possible) full data
- For SQL: folding still active on key steps
- No unexpected nulls or errors in critical columns



9. Quick Reference — Ribbon Locations

Task	Where
Use First Row as Headers	Home → Use First Row as Headers
Remove Columns	Home or Transform → Remove Columns
Remove Rows / Blanks / Duplicates	Home → Remove Rows
Change Type	Transform → Data Type (or column icon)
Replace Values / Errors	Transform → Replace Values / right-click column
Trim / Clean / Case	Transform → Format
Split Column	Transform → Split Column
Unpivot	Transform → Unpivot Columns
Fill Down/Up	Transform → Fill
Conditional Column	Add Column → Conditional Column
Custom Column	Add Column → Custom Column
Merge Queries	Home → Merge Queries
Append Queries	Home → Append Queries
Close & Apply	Home → Close & Apply

Table 2 — Common commands



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Clean early. Name steps. Refresh with confidence.

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